👩🏾‍🏫 Staff Guide · Chemistry · Unit 1A Charges & Static (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Mr. Brittle's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 1 Lightning: L1.1 Initial Model of Lightning, L1.2 Smaller Scales, L1.3 Sparks, L1.5 Particle Model of Static, L1.6 Cause of Static")	**Level 2 piece**
L1.1 Initial Model of Lightning + slides	Slides "Lightning" + notes part A
L1.2 Smaller Scales (rain/lightning graph, correlation vs causation)	Notes part A; Level 1 skips the graph
L1.3 Sparks (water dropper stations)	Static Electricity Stations lab packet + slides "Sparks"
L1.5 Particle Model of Static + L1.6 Cause of Static (reading)	Slides "Atoms", "Rubbing moves electrons" + notes parts B-C
Class slides	Deck: 19 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: ➕ proton, ➖ electron, ⚪ neutron, 🎈 rubbed balloon
- Level 1 never needs the word "nucleus" or "correlation"

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Progress tracker check-offs: use the notes in place of each L1.x worksheet
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Mr. Brittle".

6. Photo credits (open licenses)

Lightning 14.07.2009 20-42-33.JPG — Hansueli Krapf, CC BY-SA 3.0, Wikimedia Commons · Attractive-electric-force-between-hair-and-balloon.jpg — MikeRun, CC BY-SA 4.0, Wikimedia Commons

👩‍🏫 Staff Guide · Static Electricity Stations (Level 2)

Chemistry · Unit 1A · goes with the class “L1.3 Sparks, L1.5 Particle Model of Static, L1.8 Neutral Interactions” (same points as the class worksheet). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Station card**: L1.3 Sparks stations + L1.5 / L1.8 balloon activities: balloon on hair, tinsel, paper scraps, bending a stream of water. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Push or pull?”
- “Show me how far away it starts to move.”
- “What did the balloon get from your hair?”

3. Modifications

- Data table given
- Word bank for the explanation

4. What to grade (same points as the class worksheet)

- Same points: data table + one explanation sentence
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

6. Photo credits (open licenses)

Attractive-electric-force-between-hair-and-balloon.jpg — MikeRun, CC BY-SA 4.0, Wikimedia Commons

⚡ Lightning lessons 1-6 notes

Chemistry · Unit 1A Charges & Static · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Lightning and models (L1.1, L1.2)

1. Lightning is a giant electric _____.
2. Lightning makes the sound we call _____.
3. Rain and lightning happen together. This is a _____, not proof of cause.

Word bank: **correlation** **spark** **thunder**

B. Atoms (L1.5)

4. Protons have a _____ charge.
5. Electrons have a _____ charge.
6. An atom with equal + and − is _____.
7. Match the particle to where it is:

proton	nucleus, no charge
electron	outside, can move
neutron	nucleus, does not move

Word bank: **negative** **neutral** **positive**

C. Static and sparks (L1.3, L1.6)

8. Rubbing moves _____ from one object to another.
9. Extra electrons make an object _____.
10. Number the steps (1-4):

_____ Electrons move to the balloon

_____ Charge jumps: a spark

_____ Rub the balloon on hair

_____ Charge builds up

Word bank: **electrons** **negative**

Level 2 version of the gen ed “L1.1-L1.6 worksheets.”

📖 Charges & Static: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



⚡

Lightning

A giant electric spark between a cloud and the ground (or two clouds).

⚛️

Atom

The tiny particle that all matter is made of.

➕

Proton

A positive (+) particle in the center (nucleus) of an atom. It does not move.

➖

Electron

A negative (−) particle around the nucleus. Electrons can move from one object to another.

Neutron

A particle in the nucleus with no charge (neutral).



Static charge

Extra or missing electrons that build up on an object, like a rubbed balloon.

Neutral

An object with the same number of protons and electrons, so it has no overall charge.

Correlation

Two things happen together. It does not prove one causes the other.

🎈 Static Electricity Stations

Chemistry · Unit 1A · Level 2



Name: _____

L1.3 Sparks stations + L1.5 / L1.8 balloon activities: balloon on hair, tinsel, paper scraps, bending a stream of water.

🧺 At the stations

- ☐ 🎈 **1** balloon per pair
- ☐ 📄 **1 pile** tiny paper scraps
- ☐ ✨ **1** piece of tinsel
- ☐ 🚰 **1** sink or water bottle with a thin stream

🌱 Steps: check each box when you finish

1. 🎈 **Charge** the balloon: rub it on hair or wool for 10 seconds. ☐
2. 📄 **Station 1**: hold it 2 cm above paper scraps. Record what happens. ☐
3. ✨ **Station 2**: hold it near tinsel. Record attract or repel. ☐
4. 🚰 **Station 3**: hold it near a thin stream of water. Record what happens. ☐
5. 🎈🎈 **Station 4**: charge 2 balloons, hang them close. Record attract or repel. ☐
6. 📝 **Explain**: which charges were the same? Which were opposite? ☐

📋 Data table

Station	Attract, repel, or nothing?
1 Paper scraps	
2 Tinsel	
3 Water stream	
4 Two balloons	

🎈 Static Electricity Stations · Prep Sheet

Chemistry · Unit 1A · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🎈 **Rubber**	Rubs the balloon on hair or a sweater for 10 seconds.	
👀 **Watcher**	Watches closely and says what moves.	
📝 **Recorder**	Writes or circles what happened at each station.	
⏱️ **Timer**	Counts 10 seconds of rubbing and moves the group on.	

🧰 Tools and supplies: check when you have it

- ☐ 🎈 **Balloon** rub it on hair or a wool sweater
- ☐ 📄 **Paper scraps** tiny pieces
- ☐ ✨ **Tinsel** light, shiny strip
- ☐ 📋 **Station card** check each box

⚠️ Safety and manners focus

- 🎈 Do not pop balloons near faces. Tell staff about latex allergies.
- 💧 Wipe up water spills right away.
- 🔌 Stay away from outlets and computers with the charged balloon.

✅ Before you start

☐ Latex allergy check ☐ Balloon ready ☐ Paper scraps on the table

🎈 Static Electricity Stations · Reflection

Chemistry · Unit 1A · Level 2



Name: _____

📸 My photos / proof

☐ Paper scraps ☐ Bent water stream ☐ Two balloons

✅ Skills I used today (check all you did)

☐ 🎈 I charged the balloon by rubbing

☐ 👀 I observed all 4 stations

☐ 📝 I recorded attract or repel

☐ 🤝 I did my group job

⭐ Which station surprised you? Why?

1. _____
2. _____

Ideas: **paper jumped** **water bent** **balloons pushed**

🧲 What moved: electrons or protons?

1. _____
2. _____

Ideas: **electrons move** **protons stay** **rubbing**

✅ Charges & Static Quiz

Chemistry · Unit 1A · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. ⚡ Lightning is…

A. Burning gas | B. A giant electric spark | C. Sunlight

2. 🔗 Rain and lightning happen together. This is…

A. Proof that rain causes lightning | B. A charge | C. A correlation

3. ➕ A proton has…

A. A positive charge | B. A negative charge | C. No charge

4. ➖ An electron has…

A. A positive charge | B. No charge | C. A negative charge

5. ⚪ A neutron has…

A. No charge | B. A positive charge | C. A negative charge

6. ⚖️ A neutral atom has…

A. Only neutrons | B. The same number of protons and electrons | C. More electrons

7. 🎈 When a balloon is rubbed on hair, what moves?

A. Protons | B. Electrons | C. Neutrons

8. ➖ A balloon with extra electrons is…

A. Negative | B. Positive | C. Neutral

9. ✨ A spark happens when…

A. Water boils | B. Protons leave the nucleus | C. Built-up charge jumps through air

10. 🌩️ Charge separates in a storm cloud when…

A. Ice and water bump and rub | B. The cloud gets cold | C. The sun shines

⚡ Lightning lessons 1-6 notes · ANSWER KEY

Chemistry · Unit 1A Charges & Static · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Lightning and models (L1.1, L1.2)

1. Lightning is a giant electric **spark** .
2. Lightning makes the sound we call **thunder** .
3. Rain and lightning happen together. This is a **correlation** , not proof of cause.

Word bank: correlation | spark | thunder

B. Atoms (L1.5)

4. Protons have a **positive** charge.
5. Electrons have a **negative** charge.
6. An atom with equal + and − is **neutral** .
7. Match the particle to where it is:

(Key: proton → nucleus, does not move · electron → outside, can move · neutron → nucleus, no charge)

proton	nucleus, no charge
electron	outside, can move
neutron	nucleus, does not move

Word bank: negative | neutral | positive

C. Static and sparks (L1.3, L1.6)

8. Rubbing moves **electrons** from one object to another.
9. Extra electrons make an object **negative** .
10. Number the steps (1-4):

2 Electrons move to the balloon

4 Charge jumps: a spark

1 Rub the balloon on hair

3 Charge builds up

Word bank: electrons | negative

Level 2 version of the gen ed "L1.1-L1.6 worksheets."

🔑 *Key written by our team from Mr. Brittle's materials, please check.*

✅ Charges & Static Quiz · ANSWER KEY

Chemistry · Unit 1A · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. ⚡ Lightning is… L1.1

A. Burning gas | **B. A giant electric spark** | C. Sunlight

2. 🔗 Rain and lightning happen together. This is… L1.2

A. Proof that rain causes lightning | B. A charge | **C. A correlation**

3. ➕ A proton has… L1.5

A. A positive charge | B. A negative charge | C. No charge

4. ➖ An electron has… L1.5

A. A positive charge | B. No charge | **C. A negative charge**

5. ⚪ A neutron has… L1.5

A. No charge | B. A positive charge | C. A negative charge

6. ⚖️ A neutral atom has… L1.5

A. Only neutrons | **B. The same number of protons and electrons** | C. More electrons

7. 🎈 When a balloon is rubbed on hair, what moves? L1.6

A. Protons | **B. Electrons** | C. Neutrons

8. ➖ A balloon with extra electrons is… L1.6

A. Negative | B. Positive | C. Neutral

9. ✨ A spark happens when… L1.3

A. Water boils | B. Protons leave the nucleus | **C. Built-up charge jumps through air**

10. ⛈️ Charge separates in a storm cloud when… L1.6

A. Ice and water bump and rub | B. The cloud gets cold | C. The sun shines

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "L1.9 Static Interaction Review + L1.14 Practice Test (part 1)". Score: __ / 10.

🔑 *Key written by our team from Mr. Brittle's materials, please check.*